Project Documentation:

Heart Disease Analysis Dashboard

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1. Project Overview

This document details the development and implementation of the Heart Disease Analysis web application. The primary objective of this project is to leverage data visualization to uncover underlying patterns, risk factors, and demographic correlations associated with cardiovascular diseases. The final deliverable integrates interactive Tableau dashboards into a custom Flask-based web environment, providing users with a seamless analytical experience.

2. Problem Statement

Cardiovascular disease remains a leading cause of global mortality. The ability to accurately identify at-risk populations and understand the compounding effects of various lifestyle and medical factors is critical for early prevention. This project seeks to bridge the gap between raw medical data and actionable insights by visually interpreting the variables that most significantly impact heart health.

3. Data Acquisition

The foundational dataset comprises a wide array of patient attributes, encompassing demographic details (age, gender, race), lifestyle choices (smoking status, alcohol consumption), and pre-existing medical conditions (diabetes, prior strokes).

4. Data Processing and Preparation

To ensure high-fidelity visualizations, the raw dataset underwent a systematic cleaning process. This phase included:

- Identifying and resolving missing or anomalous values.
- Standardizing categorical data for accurate grouping.

- Filtering out extraneous columns to optimize processing speed.
- Converting the finalized dataset into a Tableau Extract (`.hyper`) file to meet the cloud publishing requirements of Tableau Public.

5. Visual Analytics via Tableau

The analytical core of the project was built using Tableau Desktop/Public. We developed a comprehensive suite of visual tools, organized into targeted dashboards and a narrative story, to break down complex medical correlations.

Visual Components:

- **Dashboard 1 (Demographics):** Focuses on the distribution of heart disease across different genders, racial groups, and highlights overlaps with conditions like diabetes and stroke.
- **Dashboard 2 (Health Metrics):** Provides comparative insights into specific physiological variables and their varying impacts on patient health.
- **Dashboard 3 (Lifestyle Factors):** An extended analysis isolating behavioral risks, primarily focusing on the impact of smoking and alcohol.

Project Links:

- **Interactive Story:** https://public.tableau.com/views/HeartDiseaseAnalysis-DAProject/Story?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
- **Dashboard:** https://public.tableau.com/views/HeartDiseaseAnalysis-DAProject1/Dashboard1?:language=en-US&publish=yes&:sid=&:display_count=n&:origin=viz_share_link
- **GitHub:** https://github.com/Pranjil-Singh/Heart-Disease-Analysis

6. Core Findings

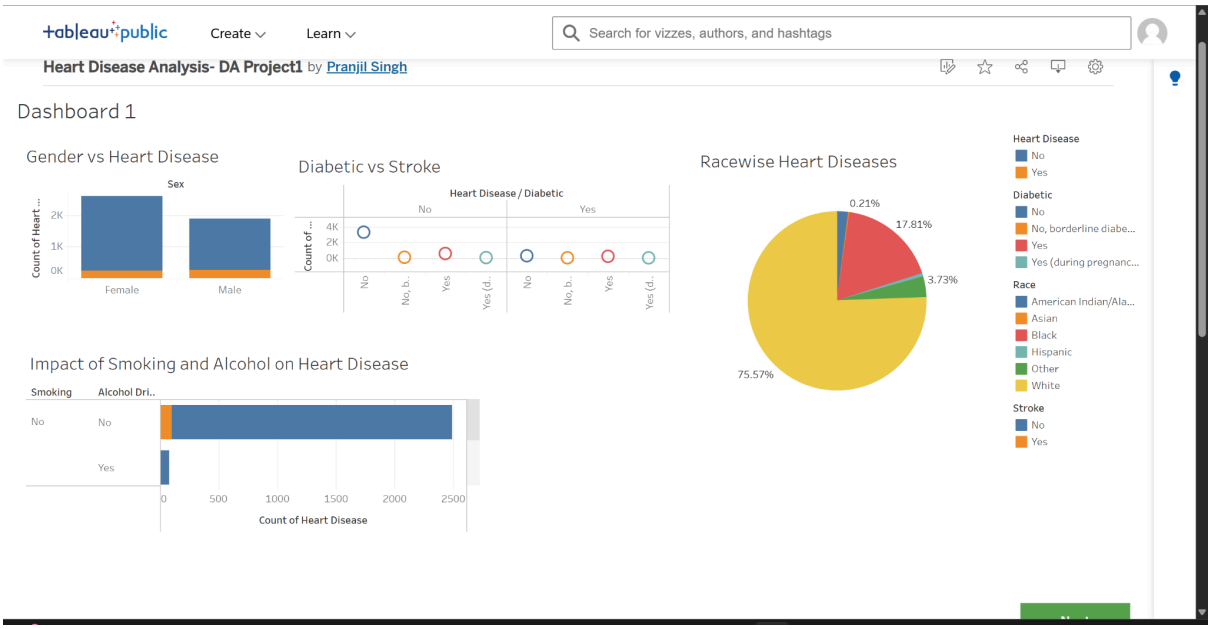
Through our visual analysis, several key trends emerged:

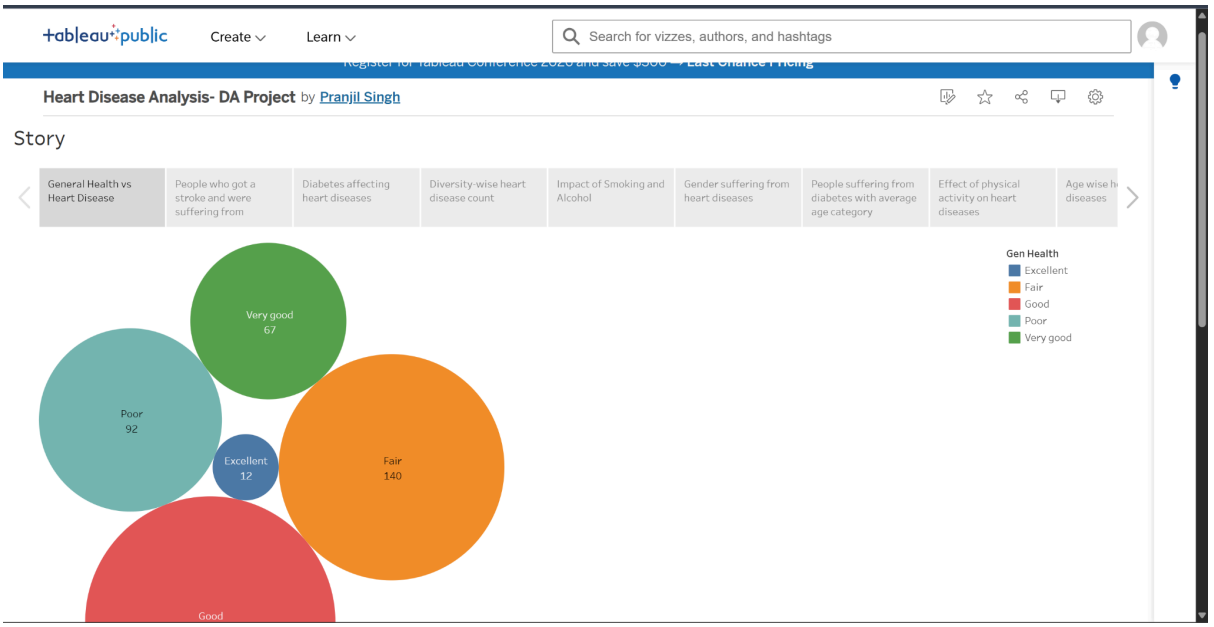
- **Age Correlation:** Advanced age serves as one of the most significant baseline predictors for heart disease onset.
- **Comorbidities:** There is a distinct, measurable overlap between patients who have experienced a stroke or severe diabetes and those diagnosed with heart disease.
- **Lifestyle Impact:** Behavioral factors, notably smoking, demonstrated a clear positive correlation with increased cardiovascular risk.
- **Demographic Variances:** Susceptibility to heart disease fluctuates across different racial and gender demographics, indicating a need for tailored preventative care.

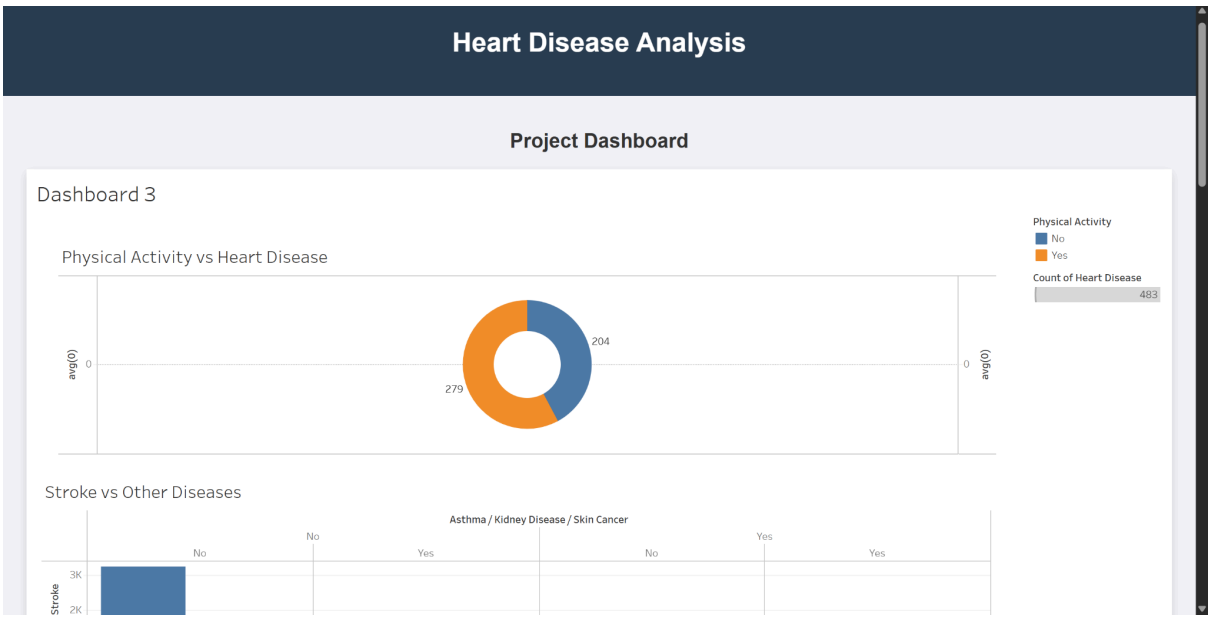
7. Application Architecture and Deployment

Rather than relying solely on the Tableau Public interface, we engineered a standalone web application to host our findings professionally. Utilizing the Tableau Embedding API v3 (Web Components), we successfully integrated responsive, interactive visualization containers directly into a custom HTML/CSS frontend, served by a Python backend.

- **Source Code Repository:** https://github.com/Pranjil-Singh/Heart-Disease-Analysis
- **Screenshots:**







8. Technology Stack

- **Data Visualization:** Tableau Public (Dashboards & Stories)
- **Backend Framework:** Python with Flask
- **Frontend Development:** HTML5, CSS3, Vanilla JavaScript
- **API Integration:** Tableau Embedding API v3 (`<tableau-viz>`)
- **Version Control:** Git and GitHub

9. Conclusion

This project successfully illustrates the power of combining data visualization with modern web development. By translating raw health data into interactive Tableau environments and deploying them via a custom Flask architecture, we created an accessible, insightful tool for analyzing the multifaceted risk factors associated with heart disease.